

Project Topic: Credit Quantity, Credit Prices, and Future Output

A model of heterogeneous financing access

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<https://github.com/amchavez/ai-project>

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1 The question and the track

This is a **Track B** project: I formulate the economic model that is currently missing from my undergraduate thesis. The question is: *when does the quantity of bank credit contain more information about firms' future production than the price of credit, and can heterogeneous access to outside finance explain why those margins may differ across sectors?*

The thesis is still in progress. Its preliminary forecasting estimates suggest that credit quantities may be more informative for non-tradable activity at relatively short horizons, while a credit-price indicator may become more informative for tradable activity over a longer horizon. I treat this as a qualitative pattern to motivate the model, not as a settled empirical fact or a causal estimate. The project asks for a mechanism that can generate the cross-sector ranking before trying to explain its exact timing.

2 The baseline and the assumption I add

The empirical thesis compares autoregressive forecasts with and without credit indicators. That exercise can establish incremental predictive content, but it does not contain a firm problem that distinguishes the quantity of funds from their marginal price. I add firms that differ in their access to alternative finance.

There are two dates. Firm i enters date zero with internal funds $n_i > 0$, chooses capital k_i , bank borrowing b_i , and outside finance m_i , and produces $F_i(k_i)$ at date one. It solves

$$\max_{k_i, b_i, m_i} \beta F_i(k_i) - R_B b_i - R_M m_i \quad \text{s.t.} \quad k_i = n_i + b_i + m_i, \quad 0 \leq b_i \leq \bar{b}_i, \quad m_i \geq 0. \quad (1)$$

For a bank-dependent firm, $m_i = 0$. A firm with market access may use $m_i > 0$ after exhausting its bank line. The maintained conditions are

$$F'_i(k) > 0, \quad F''_i(k) < 0, \\ 0 < R_B \leq R_M = R_B + \tau_i, \quad \tau_i \geq 0.$$

The financing order means that the firm uses internal funds first, bank credit second, and outside finance last. The bank line creates a quantity margin; R_M creates a marginal-price margin. For closed forms and formal verification I use

$$F_i(k) = a_i k - \frac{c_i}{2} k^2, \quad a_i, c_i > 0, \quad (2)$$

on its increasing region. Let $\bar{k}_i = n_i + \bar{b}_i$ denote capital at the bank ceiling. This is a deliberately small partial-equilibrium model. It takes the credit line and finance wedges as given rather than claiming to identify the source of a credit shock.

3 The result I expect

The left and right marginal conditions are

$$\beta F'_i(k) = R_B \quad \text{and} \quad \beta F'_i(k) = R_M. \quad (3)$$

Under equation (2), the corresponding targets are

$$k_B = \frac{\beta a_i - R_B}{\beta c_i}, \quad k_M = \frac{\beta a_i - R_M}{\beta c_i}, \quad k_M \leq k_B.$$

I expect the market-access firm's policy to be

$$k_i^* = \begin{cases} n_i, & k_B \leq n_i, \\ k_B, & n_i < k_B < \bar{k}_i, \\ \bar{k}_i, & k_M \leq \bar{k}_i \leq k_B, \\ k_M, & \bar{k}_i < k_M. \end{cases} \quad (4)$$

The main conjecture compares the last two economically relevant cases, holding the regime fixed under a small parameter change. If $k_M < \bar{k}_i < k_B$, the bank ceiling binds at a kink and

$$\frac{\partial k_i^*}{\partial \bar{b}_i} = 1, \quad \frac{\partial F_i(k_i^*)}{\partial \bar{b}_i} = F'_i(\bar{k}_i) > 0.$$

The available quantity of bank finance therefore determines the scale of the project. A small common spread movement has no local real effect while the firm stays at the kink.

If $\bar{k}_i < k_M$, outside finance funds the marginal unit. A larger bank line then replaces outside finance without changing total capital, while a common credit-cost factor z that raises both R_B and R_M gives

$$\frac{\partial k_i^*}{\partial \bar{b}_i} = 0, \quad \frac{\partial k_i^*}{\partial z} = \frac{1}{\beta F''_i(k_M)} < 0.$$

The spread matters because it changes the cost of the marginal unit, not merely because it correlates with the business cycle.

Let λ_s be the share of firms in sector s at the quantity-sensitive kink. If $\lambda_N > \lambda_T$ and marginal products are positive, then the non-tradable sector is more sensitive to the credit ceiling and the tradable sector is more sensitive, in absolute value, to the common spread. This is the qualitative bridge to the preliminary thesis pattern. Slow-moving financing access can make current credit quantities reveal the near-term investment pipeline of constrained firms. Credit prices can contain information about the desired scale of projects for firms that can substitute across funding sources. Different adjustment speeds may move the peak response across horizons, but the baseline model does not yet derive exact forecast horizons.

4 Why this is not already solved

I compared the published articles, publicly available working-paper versions, and appendices of the closest contributions. [Holmström and Tirole \(1997\)](#) derive how firm and intermediary capital determine monitored and direct finance. Their mechanism motivates heterogeneous access, but it does not rank credit quantities and prices as sector-specific forecasting indicators. [Gilchrist and Zakrajšek \(2012\)](#) show that corporate-bond spreads predict aggregate activity and decompose the spread into expected-default and excess premium components. Their working-paper forecasting exercises cover short and longer horizons, but they do not compare that price signal with a binding bank quantity at the firm level.

[Müller and Verner \(2024\)](#) are the closest sectoral paper. Their published *Review of Economic Studies* article and its NBER working-paper appendix show that non-tradable credit expansions have different macroeconomic content from tradable credit expansions and examine bond issuance as a robustness check. Their object is the allocation of credit across sectors, not the choice between a capped bank tranche and a more expensive marginal source. Follow-on work such as [Gilchrist et al. \(2014\)](#) models how spreads affect firm investment, but does not deliver the quantity-versus-price sectoral ranking. The proposed contribution is therefore narrow: a transparent threshold model tailored to the thesis’s two predictors. I do not claim that credit rationing, outside finance, or sectoral heterogeneity is new.

5 Plan and risks

The next steps are to (i) prove the complete piecewise policy and both sectoral comparative statics; (ii) make `code/verify.py` reproduce every reported number and fail outside the stated conditions; (iii) test behavior close to the regime boundaries; and (iv) formalize every numbered result in Lean using the course’s AppliedModelingLib workflow. A dynamic extension will add capital accumulation, adjustment costs, and the shadow value of the credit limit, but it will remain separate from the baseline proposition until its conditions can be proved.

The main empirical risk is that the thesis data do not directly measure firms’ outside-finance access, so the sectoral composition condition $\lambda_N > \lambda_T$ may remain a plausible hypothesis rather than an identified fact. The main theoretical risk is that the sharp local zero in the kink regime disappears once the credit ceiling or funding costs are jointly endogenous. If either risk materializes, I will keep the result conditional, report regime switching in simulations, and avoid using the model to claim exact timing or causality. The Lean risk is the piecewise optimum; the fallback is to formalize the quadratic specialization first and label any remaining general concave result as open.

References

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